INCH-POUND
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DETAIL SPECIFICATION

PALLETS, CARGO, AIRCRAFT, TYPE HCU-6/E

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.

This specification covers, aircraft cargo pallets, with a load capacity of 10,000 pounds.

2. APPLICABLE DOCUMENTS

2.1 General.

The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to AFLCMC/EZSS, 2145 Monahan Way, Bldg. 28, Wright-Patterson AFB OH, 45433-7017 or e-mailed to Engineering.Standards@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-61002	Labels, Pressure-Sensitive Adhesive, for Bar Coding

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	Identification Marking of U.S. Military Property Ground and Ground Support Equipment
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-889	Dissimilar Metals
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft

DEPARTMENT OF DEFENSE HANDBOOK

MIL-HDBK-831	Preparation of Test Reports

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS - AIR FORCE

201993426	Pallet Assembly, Cargo – HCU-6A/E, Next-Gen
201993431	Tie-Down Ring Assembly - Pallet, Airborne Cargo

(Copies of these drawings are available online at https://jedmics.af.mil/webjedmics/index.jsp, for users with approved access rights granted through the Joint Engineering Data Management Information and Control System (JEDMICS). Users not located on a *.mil network may contact the JEDMICS Help Desk via email at JEDMICS@robins.af.mil for instructions on obtaining and completing the necessary forms.)

2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4	Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted)

(Copies of this document are available online at http://www.asq.org/.)

ASTM INTERNATIONAL

ASTM D6251/D6251M	Standard Specification for Wood-Cleated Panelboard Shipping Boxes

(Copies of this document are available online at http://www.astm.org/.)

AMERICAN WELDING SOCIETY (AWS)

AWS B4.0	Standard Methods for Mechanical Testing of Welds
AWS D1.2/D1.2M	Structural Welding Code - Aluminum
AWS D17.3/D17.3M	Specification for Friction Stir Welding of Aluminum Alloys for Aerospace Applications

(Copies of these documents are available online at http://www.aws.org/.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 8256	Plastics - Determination of Tensile-Impact Strength

(Copies of this document are available online at http://www.iso.org/.)

SAE INTERNATIONAL

SAE-AMS-STD-595	Colors Used in Government Procurement

(Copies of this document are available online at http://www.sae.org/.)

2.4 Order of precedence.

Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article.

When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

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3.2 Materials.

3.2.1 Protective treatment.
The materials shall be as specified by applicable drawings and details thereof. When materials are used in the construction of pallets that are subject to deterioration when exposed to climatic and environmental conditions that are likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with performance requirements of this specification (see 4.5.1). The use of any protective coating that will crack, chip, or scale with age or extreme climatic and environmental conditions shall be avoided.

3.2.2 Metals.
Metals shall be of the corrosion-resistant type or treated to resist corrosion due to salt spray or atmospheric conditions likely to be met in storage or normal service (see 4.5.1).

3.2.2.1 Dissimilar metals.
Contact between dissimilar metals shall be avoided wherever possible (see 4.5.1). Wherever such contacts are unavoidable, they shall be insulated in a manner approved by the procuring activity. Dissimilar metals are defined in MIL-STD-889.

3.3 Construction.
The pallet shall be constructed so that no parts shall work loose in service. It shall be built to withstand the strains, jars, vibration, and other conditions incident to shipping, storage, installation, and service. The pallet shall be constructed as an integral unit and shall not require assembly or attachment of separate parts during operational use. The pallet shall be constructed and assembled in accordance with this specification and AF Drawing 201993426.

3.3.1 Reliability.
The pallet shall be constructed as specified herein (see 4.5.1) to assure optimum service life under all service conditions specified herein, and shall be acceptably demonstrated during the testing specified herein. The pallet shall be designed for life of a minimum of 10 years.

3.3.2 Load center of gravity.
The vertical center-of-gravity shall be 48 inches above pallet surface. Load configurations (see 3.3.3) with alternative center-of-gravity heights may be used for all tests except the Ultimate Load (3.4.6). The loads shall be uniformly distributed for each type of pallet.

3.3.3 Load configuration.
The load for each pallet shall be rectangular cubes (boxes). The boxes shall be constructed in accordance with ASTM D6251/D6251M. The cubes for pallets shall be approximately 20 × 20 × 12 inches. All cubes shall be free to move relative to adjacent cubes. Alternative load configurations may be used for all tests except the Ultimate Load (see 3.4.5). These alternative load configurations (e.g., steel or concrete blocks) must load the entire surface of the pallet evenly. Furthermore, alternative load configurations must not include continuous steel sheets or other reinforcements between the pallet and the load.

3.3.4 Welds.
Welds shall be continuous without gaps, cracks, or missing areas and shall be per the applicable

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drawing. The supplier shall conduct process development and qualitative and quantitative testing to demonstrate compliance with AWS D17.3/D17.3M, Class B for friction stir welds (FSW) (dye penetrant inspection not required) and to demonstrate acceptable weld quality and weld efficiency for both gas metal arc welding (GMAW) and gas tungsten arc welding (GTAW) (see 4.4.2.4 and 4.5.9.2) in accordance with AWS D1.2/D1.2M. For GMAW/GTAW specimens, the average weld efficiency (weld specimen failure stress/parent extruded base material specimen failure stress) × 100 percent shall be greater than 33 percent. For FSW specimens, the weld efficiency (ultimate tensile strength of the welded test specimen / specified minimum tensile strength of the parent extruded base material) × 100 percent shall be greater than 60 percent. Note that defect and strength requirements may be relaxed if quantitative testing is conducted and approved by the government demonstrating that the proposed requirements do not result in strength reductions below those specified and do not affect other critical test capability (see 4.5.3.4). The supplier shall also develop a Welding Procedure Specification (WPS) and maintain Procedure Qualification Records (PQRs) for all welding processes.

3.3.5 Weld repair.
Weld repair is allowed to recover from inadvertent weld process stop or visually detected flaw. Repair techniques allowed are FSW and GMAW/GTAW. This includes a single pass re-weld of any joint with FSW or GMAW/GTAW, utilizing a qualified WPS. The development of the repair procedures shall be qualified according to the performance requirements of AWS D17.3/D17.3M for FSW and AWS D1.2/D1.2M for GMAW/GTAW, to the extent specified in the results section of PQR. Repair criteria shall be in associated manufacturing plan or separate repair procedure. For GMAW/GTAW repairs, dye penetrant inspection is required post-repair. Repair area shall be ground as necessary to meet flushness requirements.

3.4 Performance.

3.4.1 Tie-down rings.

3.4.1.1 Ring location and movement.
Tie-down rings shall be provided according to the requirements herein and applicable AF Drawing 201993431. When not used for tie-down, the rings shall not interfere with the entry of the pallet into a rail section. The pivot point of the ring shall be outside the usable surface edge. Each ring shall be capable of at least 225 degrees of free movement in a vertical plane that intersects the pallet edge at right angles (see 4.5.3.1).

3.4.1.2 Ring capacity one.
Each ring shall be designed to sustain a 7,500 pound tension load along any line in a vertical plane that intersects the pallet edge at right angles within the angular limits (see 4.5.3.2). The pallet shall withstand loading without pallet or weld cracking and without tie-down ring loosening or rivet failure. Minor permanent deformation of the ring is permitted, as long as the ring retains full range-of-motion and the functionality of the ring is not impacted. The rings shall withstand 7,500 pounds × 1.5 (Safety Factor) = 11,250 pounds ultimate load without breaking the ring or rivets.

3.4.1.3 Ring capacity two.
Each ring shall also be designed to sustain a 7,500 pound load along any line originating at the attachment point in a vertical plane extending above the pallet and parallel to the edge (see 4.5.3.3).

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The pallet shall withstand loading without pallet or weld cracking and without tie-down ring loosening or rivet failure. Minor permanent deformation of the ring is permitted, as long as the ring retains full range-of-motion and the functionality of the ring is not impacted.

3.4.1.4 Ring capacity three.
The rings shall be designed such that two adjacent corner rings can be pulled simultaneously and then two center rings pulled simultaneously to a load of 7,500 pounds (see 4.5.3.4). The pallet shall withstand loading without pallet or weld cracking and without tie-down ring loosening or rivet failure. Minor permanent deformation of the ring is permitted, as long as the ring retains full range-of-motion and the functionality of the ring is not impacted.

3.4.2 Static load.
When supported by four rows of conveyors as specified (see 3.4.5), the pallet shall be capable of supporting a uniformly distributed static load (see 6.3.2) for 40 hours without permanent deformation (must pass flatness requirement after test, flatness test, (see 3.5.1 and 4.5.1.1)) or other damage (see 4.5.4).

3.4.3 Lifting capabilities.

3.4.3.1 Corner lifting.
When uniformly loaded (see 6.3.3), the pallet shall be capable of withstanding, without permanent deformation or other damage, lifting by means of four cables attached to the four tie-down rings adjacent to the four corners (see 4.5.5.1). Two separate tests shall be conducted: 1) the pallet shall be lifted by the corners tie-down rings on the long side and 2) the pallet shall be lifted by the corner tie-down rings on the short side.

3.4.3.2 Forklift lifting.
When uniformly loaded to rated capacity (see 6.3.1), a pallet shall withstand lifting from a conveyor without permanent deformation or other damage. The pallet shall be supported by conveyors in any of the three geometries as specified (see 3.4.4.1). A forklift truck having 72 inch long × 6 to 8 inch wide tines spaced 42 inches center-to-center shall lift the pallets. Forklift truck tines shall have an end radius of not less than 0.250 inch (see 4.5.5.2).

3.4.3.3 Forklift "wedge" lifting.
When uniformly loaded to rated capacity (see 6.3.1), a pallet shall withstand lifting from a smooth, dry, concrete surface without permanent deformation or other damage. The tines shall be wedged under the pallet perpendicular to a pallet rail and lifted just using the lip of the pallet. A forklift truck having 72 inch long × 6 to 8 inch wide tines spaced 42 inches center-to-center shall lift the pallets. Forklift truck tines shall have an end radius of not less than 0.250 inch (see 4.5.5.3).

3.4.4 Traversable capabilities.

3.4.4.1 Conveyor.
The pallet shall be uniformly loaded to each of the four distributed loads shown in Table I. The pallet shall be capable of traversing roller conveyors that are made up of the highline dock, the Tunner, and an aerial port warehouse (see Figure 1). The configurations of the three conveyor systems are to have roller row spacing and roller dimensions representative of the actual highline

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dock, Tunner, and aerial port warehouse conveyor rollers. The pallet shall be capable of performing this testing without permanent deformation (must pass flatness requirement after test, flatness test, (see 3.5.1 and 4.5.1.1)) or other damage.

TABLE I. Pallet load/distance requirements.

Pallet Load (lbs.)	Distance (ft.) parallel to 108" dimension	Distance (ft.) parallel to 88" dimension
5,000	10,000	10,000
10,000	5,000	5,000
Total	15,000	15,000

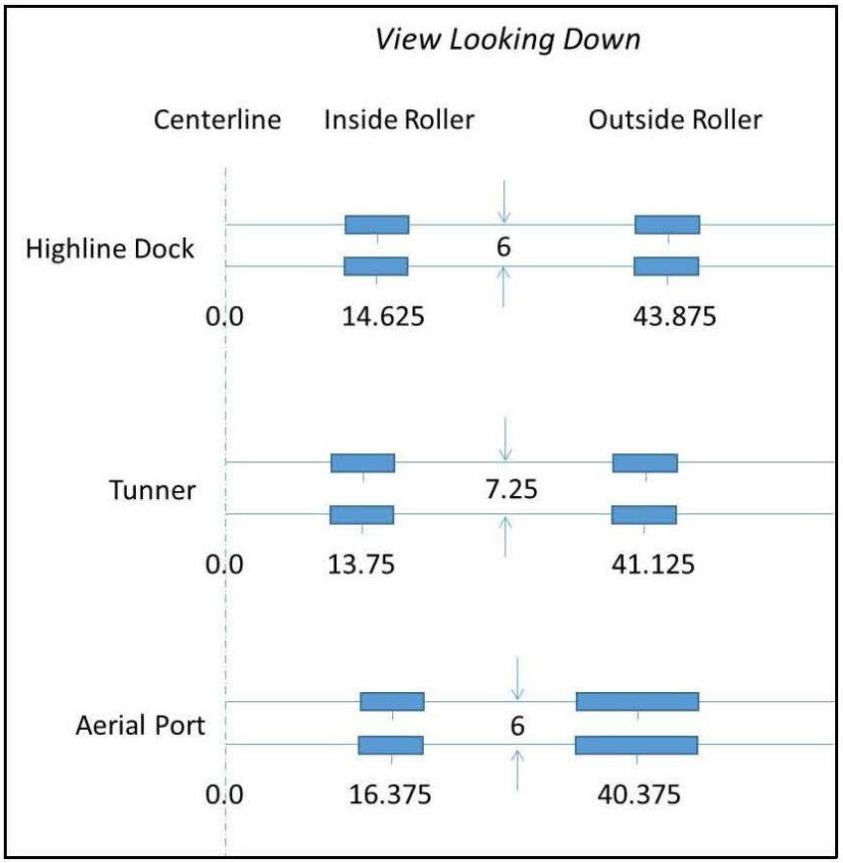


FIGURE 1. Roller geometry for conveyor test (1/3 of total length made up of each).

3.4.4.2 Conveyor ramp.
When uniformly loaded to rated capacity (see 6.3.1), the pallet shall withstand being towed over a 17 degree, 10-foot long ramp connected to a single horizontal conveyor surface above the ramp. The pallet shall be supported by conveyors in any of the three geometries as specified (see 3.4.4.1). The pallet shall be towed parallel to both 88 and 108-inch sides for Next-Gen pallets (see 4.5.6.2). The pallet shall be capable of performing this testing without permanent deformation (must pass flatness requirement after test, flatness test, (see 3.5.1 and 4.5.1.1)), or other damage.

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3.4.5 Ultimate load.
When uniformly loaded to rated capacity (see 6.3.1), the load being restrained to the pallet by nets (see 6.3.4), the pallet installed between restraining rails locked to the rails by two locks through each rail and engaging two lock notches on each side of the pallet. The pallet shall be supported by four rows of conveyors, one row along each side, and one row along a line 10 inches each side of the center line. The rollers shall be 2-inch diameter × 3.5 inch long rollers spaced on 10 inch centers. The loaded pallet shall withstand a dynamic load of three times the force of gravity (g) for a period of 0.1 second. The pallet shall be serviceable after undergoing the test. In addition, the pallet shall withstand a dynamic load of 8 g's for a period not less than 0.1 second. The pallet need not be serviceable after undergoing such a load; however, the pallet shall remain in one piece (see 4.5.7).

3.4.6 Environmental conditions.
The pallet shall be capable of operating in the following conditions without permanent damage:

a. Temperatures ranging from -65 to +165 °F (see 4.5.8.1 and 4.5.8.2).
b. Humidity (see 4.5.8.3).
c. Salt fog test (see.4.5.8.4).

3.4.7 Mechanical properties.

3.4.7.1 Pressure/Point load.
The surface of the pallet shall withstand 900 psi point loads (over one square inch) and in a location midway between the extrusion ribs, without permanent deformation (see 4.5.9.1).

3.5 Dimensions and weight.
The nominal dimensions of the Next-Gen pallet are specified on AF Drawing 201993426 (see 4.5.1). The Next-Gen pallet maximum allowable weight is 300 pounds.

3.5.1 Flatness.
The top and bottom surfaces of the pallet must be flat within 0.1875 inch (see 4.5.1.1).

3.6 Assembly.
Pallets shall be assembled according to applicable portions of AF Drawing 201993426 (see 4.5.1).

3.7 Finishes and protective coatings.
The finishes and protective coatings shall be as specified by the applicable drawings (see 4.5.1).

3.8 Identification of product.
Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. Each pallet shall have "Property of US Air Force" centered and stenciled in 1.00-inch high letters on the top of the locking lips on each long (108 inches) rail. The markings shall be made with manufacturer's standard commercial paint closest to color number 17038 of SAE-AMS-STD-595 (see 4.5.1).

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3.8.1 Identification label.
Two identification labels, which conform to MIL-PRF-61002, classification M61002-VB1b, and are marked in accordance with MIL-STD-130, shall be attached to the pallet (see 4.5.1). The identification labels shall be of a size not less than 0.750 inch × 5.00 inches and not more than 1.00 inch × 6.00 inches. One decal shall be located above the locking lips on the vertical surface 6 inches from the mitered end of the short rail; the other shall be located above the locking lips on the other short rail such that it is located at the opposite corner of the pallet (diagonally). The identification labels shall contain the following information:

a. Manufacturer's Federal Supply Code
b. Manufacturer's Serial Number
c. Contract Number
d. Date Of Manufacture
e. Date of termination of warranty
f. Bar Code (see 3.8.1.1)

3.8.1.1 Bar code.
The production facility shall include bar codes on each pallet Identification Label (see 3.8.1). The bar codes shall use code 39 format, shall be at least 0.625 inch tall, and shall have no less than six characters. At time of contract award, supplier shall furnish sequential serial numbers to equal quantities in releases against contract. These sequential serial numbers will be different for each pallet. The supplier shall also assign lot numbers. The serial and lot numbers shall be approved by WR-ALC/WNZ prior to delivery.

3.9 Painting and marking.
Each pallet shall have "Property of U. S. Air Force - Return to the nearest Air Force Base", stenciled in 1-inch high block letters on the locking lips on both long rails (see 4.5.1). Clean surface with an industrial cleaner (e.g. Skykleen 1000). The markings shall be made with the manufacturer's standard commercial paint closest to color number 17038 of SAE-AMS-STD-595.

3.10 Workmanship.
Workmanship shall be of the highest quality to ensure proper functioning under conditions to which the pallet may be subjected. Unsatisfactory workmanship, such as loose, cocked, or inadequately headed rivets; distorted or loose bushings and pins; or rough, malformed, misaligned, or improperly fabricated fittings shall be considered as defects. All corners shall be rounded, sharp edges, burrs, or protrusions shall not be permitted. Tie-down ring attaching hardware shall be positively attached to the pallet rail. Pallets shall be prepared and welded in accordance with specified procedures (see 4.5.1).

3.11 Government furnished property.
The following property will be furnished by the Government (see 6.4) to be used in testing pallets. These items shall be furnished in as close to new condition as possible.

12 each HCU-7/E or HCU-7A/E side nets and 6 each HCU-15/C top nets.

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4. VERIFICATION

4.1 Classification of inspections.

The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Test rejection criteria.

Throughout all tests specified herein, the pallet shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein.

b. Structural failure of any component, including permanent deformation that impacts functionality, cracking, or evidence of impending failure.

c. Evidence of excessive wear.

d. Misalignment of components.

e. Conditions which present a safety hazard to personnel during operation, service or maintenance.

4.3 First article testing.

4.3.1 Test samples.

The first article test sample shall consist of six pallets representative of the production pallets. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.3.1 First article tests.

The preproduction tests shall consist of all tests described under 4.5 and shall be conducted on pallets in the following sequence (**NOTE:** not all tests shown are performed on each pallet):

Samples No. 1-2	Samples No. 3-6
Examination of product (4.5.1). Tie-down ring test (4.5.3.1 through 4.5.3.4). Lifting capabilities tests (4.5.5.1 through 4.5.5.3). Environmental tests (4.5.8). Destruction analysis (4.5.9).	Examination of product (4.5.1). Static load test (4.5.4). Traversable capabilities tests (4.5.6). Ultimate load test (4.5.7).

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4.4 Quality conformance inspection.

4.4.1 Individual test.
Pallets shall be tested as specified (see 4.5.1).

4.4.2 Sampling test.

4.4.2.1 Lot.
A lot shall consist of pallets manufactured under the same conditions and submitted for inspection at the same time.

4.4.2.2 Sampling.
Sampling for examination and tests shall be in accordance with ANSI/ASQ Z1.4, General Inspection Level II, single sampling plan for normal inspection. AQL's of zero for critical, 1.5 percent for major, and 4 percent for minor.

4.4.2.3 Examination.
Samples selected in accordance with 4.4.2.2 shall be examined as specified (see 4.4.2.5).

4.4.2.4 Tests.
Samples selected in accordance with 4.4.2.2 shall be tested as specified in 4.5.3.4, 4.5.5.1, and 4.5.9.2. The supplier shall present Statistical Process Control data, as well as qualitative and quantitative test data to demonstrate compliance with AWS D17.3/D17.3M, Class B for FSW during both process development and production (dye penetrant inspection not required). Samples to demonstrate compliance with 4.4.2.2 may include off-cuts, extractions, or witnesses. GMAW and GTAW quality for corner welds verified by: 1) 100 percent visual inspection; 2) sampling per 4.4.2.2 using off-cuts, extractions, or witnesses to verify weld integrity, as well as 4.5.3.4, and 4.5.9.2.

Visual inspection personnel shall be qualified for the purposes of 4.4.2.2 (local qualification acceptable for normal production weld inspection). The basis of inspector qualification shall be documented. If the engineer elects to specify the basis of Inspector qualification, it shall be stated in the contract documents. Examples include current or previous certification as an AWS Certified Welding Inspector (CWI), Canadian Welding Bureau Inspector or as an engineer or technician who by training, or experience or both, in metals fabrication, inspection, and testing, is competent to perform inspection work.

4.4.2.5 Classification of defects.
All dimensional characteristics are considered defective when out of tolerance. The classification of defects for pallets shall be as follows:

Critical	Major	Minor
Length, width, and lip dimensions out of tolerance.	Lock notches out of tolerance.	Cocked rivet.
Pallet not processed in accordance with approved process specification.	Tie-down ring fastening device not coated with wet primer when installed.	Weight exceeds maximum.

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Failed lift test.	Tie-down ring missing.	
Flatness out of tolerance.	Tie-down ring mounting hardware loose or located incorrectly.	
Weld gaps, weld cracks, or missing weld areas.	Rivet missing.	
	Wrong rivets.	
	Tie-down rings assembly parts not plated in accordance with drawing.	
	Part not marked in accordance with MIL-STD-130.	
	Thickness exceeds tolerance.	
	Tie-down ring not coated as specified after installation.	

4.4.2.6 Rejection and retest.
When a specified sample (see 4.4.2.2) from a lot fails to meet the specification, acceptance of all pallets in the lot shall be withheld until the extent and cause of failure have been determined. The contractor shall fully explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction of the lot, all of the sampling tests shall be repeated on one pallet from this lot.

4.4.2.6.1 Individual tests may continue.
For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. However, final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

4.4.2.6.2 Defects in pallets already accepted.
The investigation of a test failure could indicate that defects may exist in pallets already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.4.3 Rejection and retest.
When a pallet fails to meet the requirements of this specification, pallets to be delivered pursuant to the contract may not be accepted until the cause of the failure is determined, corrective action is taken by the contractor, and the components that failed are successfully retested to assure compliance with this specification. Tests not involving the failed component may be continued pending the investigation of a test failure.

4.5 Test methods.

4.5.1 Examination of product.
The pallet shall be inspected to determine compliance with all requirements specified herein with respect to materials, dimensions, workmanship, welds, and marking. For the first article and random sample pallets this examination shall be accomplished by use of a checklist which lists each requirement not validated by a test and the results of the examination.

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4.5.1.1 Flatness.
Use a 108 inch minimum straight edge and 0.1875 inch gauge. Flatness is defined as A or B on Figure 2 depending on whether the surface is generally concave or convex. Note that weld beads or friction stir weld flashing shall not be included in this measurement. Check pallet at each location shown by dotted lines on Figure 2.

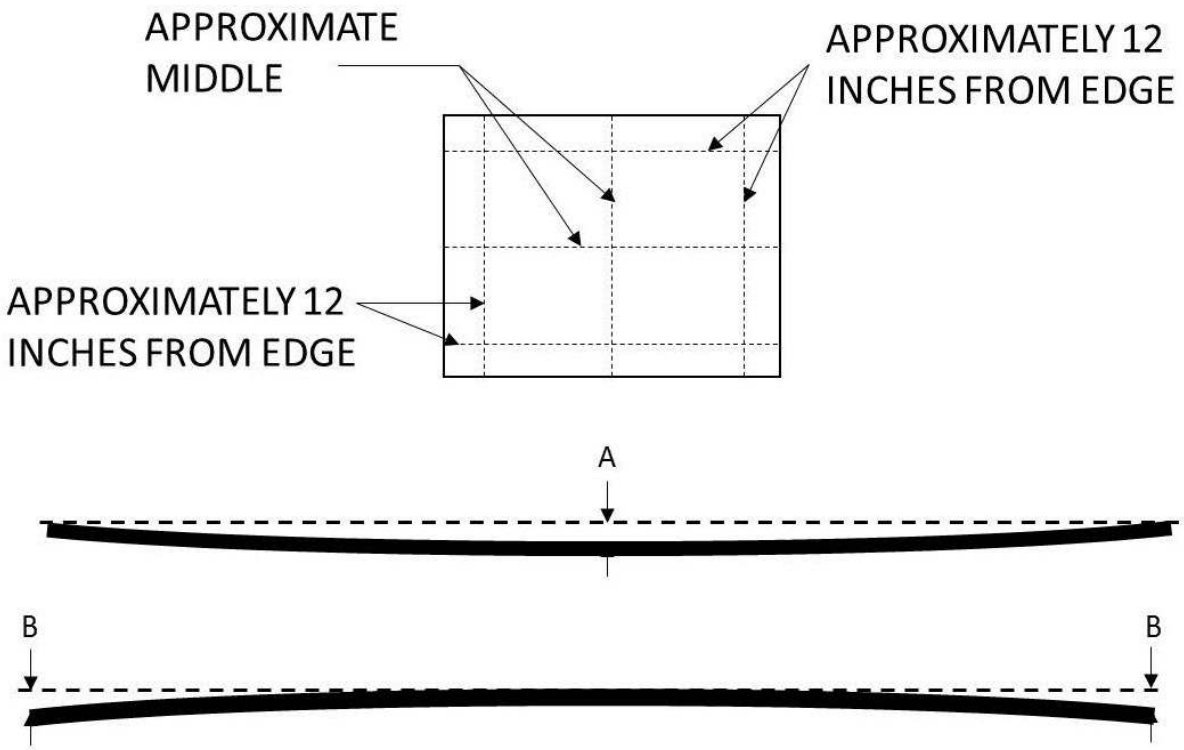


FIGURE 2. Flatness test.

4.5.2 Tap test.
Not applicable.

4.5.3 Tie-down ring tests.
Tie-down ring testing shall be conducted with a clevis or hook with a minimum cross sectional area of 0.875 square inch. The tolerance on loading angles is ±2 degrees. The pallet shall withstand each loading condition without permanent deformation, pallet or weld cracking, and without tie-down ring damage, loosening, or rivet failure, with the exception of the ultimate load testing in 4.5.3.2. Minor permanent deformation of the ring is allowable, as long as the ring retains full range-of-motion and the functionality of the ring is not impacted.

4.5.3.1 Ring location and movement test.
Compliance with the ring location and free movement requirements (see 3.4.1.1) shall be demonstrated for each ring.

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4.5.3.2 Ring load test one.
Two tie-down rings, one located at the pallet corner and the other located midway on the long side, shall be subjected to loading tests. In a plane that intersects the attachment point and is perpendicular to the pallet surface and edge, the ring shall be sequentially loaded to 7,500 pounds along lines that form 30, 45, 90, 135, and 180 degrees with the upper pallet surface (see 3.4.1.2), (see Figure 3). After the ring successfully passes these tests, it shall be loaded to 11,250 pounds (ultimate load = 7,500 × 1.5 = 11,250 pounds) along lines that form 90 and 180 degrees with the upper pallet surface. 90 and 180 degrees represent the worst-case orientation for the rings, since the rivets are in either pure shear or pure tension. For this ultimate load test, deformation of the tie-down ring assembly and rivet loosening are acceptable as long as the tie-down ring assembly remains attached to the pallet. Note that this 11,250 pound load case may be conducted using the test specimen from 4.5.8.4 to avoid damaging a serviceable pallet. The pallet (see 4.5.8.4) shall be clamped down to a rigid surface for this testing. Loading of the ring may use a hook or clevis, with a load cell between the loading device and the hook or clevis with a minimum cross sectional area of 0.875 square inch.

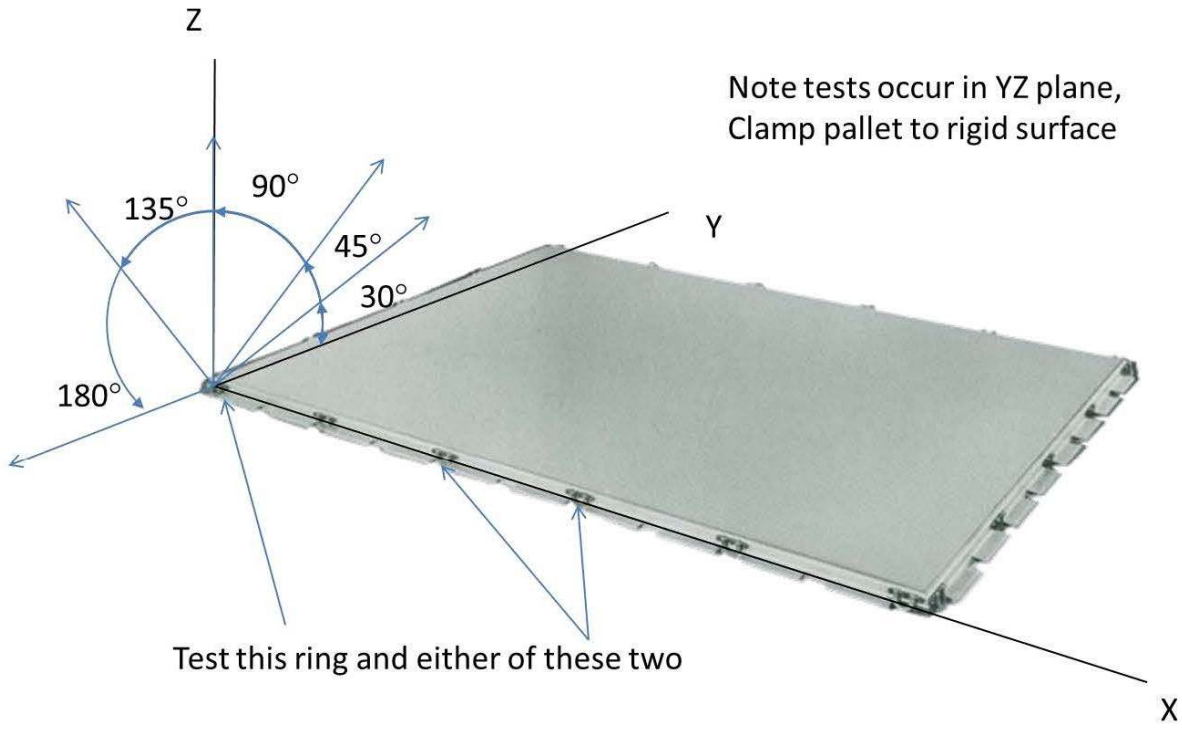


FIGURE 3. Ring tie-down test one (isometric and side views).

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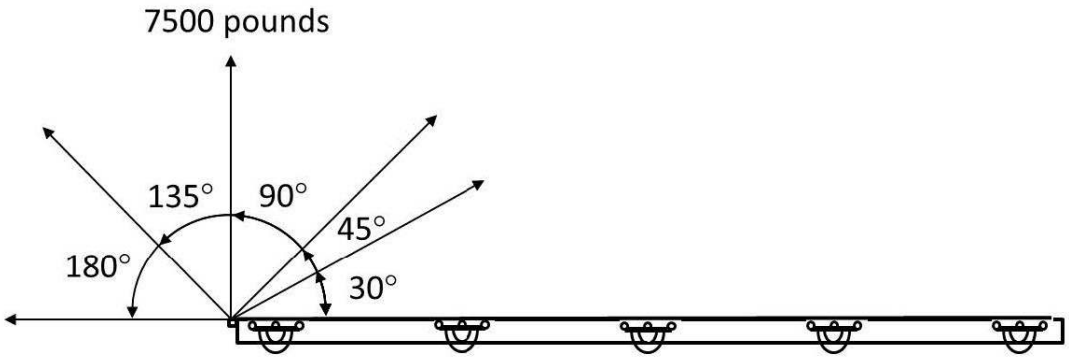


FIGURE 3. Ring tie-down test one (isometric and side views) - Continued.

4.4.5.3 Ring load test two.
In a vertical plane that intersects the attachment point and is parallel to the pallet edge, a 7,500-pound load shall be applied sequentially in two directions parallel to the upper pallet surface and at 45 and 135 degree angles with this surface (see 3.4.1.3 and Figure 4).

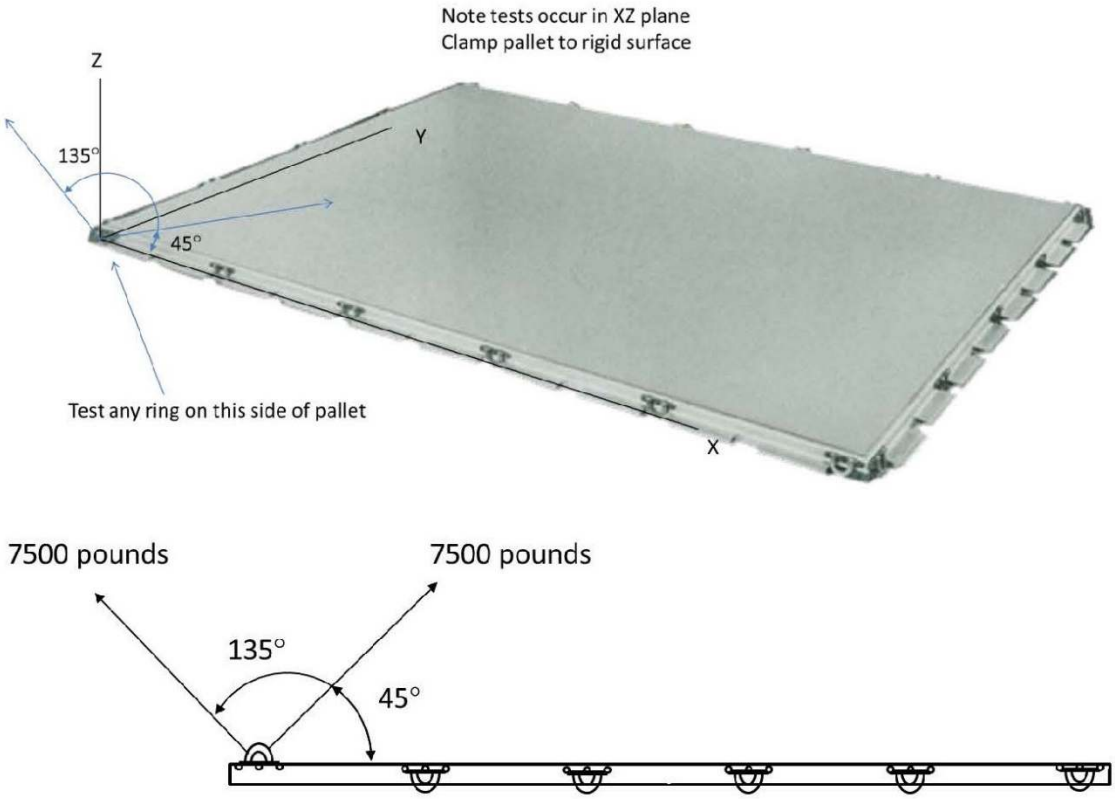


FIGURE 4. Ring tie-down test two (isometric and side views).

4.5.3.4 Ring load test three.
The pallet shall be placed in a horizontal plane and two adjacent corner rings pulled simultaneously

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and then two center rings pulled simultaneously, (see Figure 5). The load shall be 7,500 pounds. At the completion of the tests any permanent deformation or cracking of the pallet, or tie-down ring loosening, rivet failure, or substantial permanent deformation that impacts functionality, shall be cause for rejection. The load during each test shall be applied for not less than three times each (see 3.4.1.4).

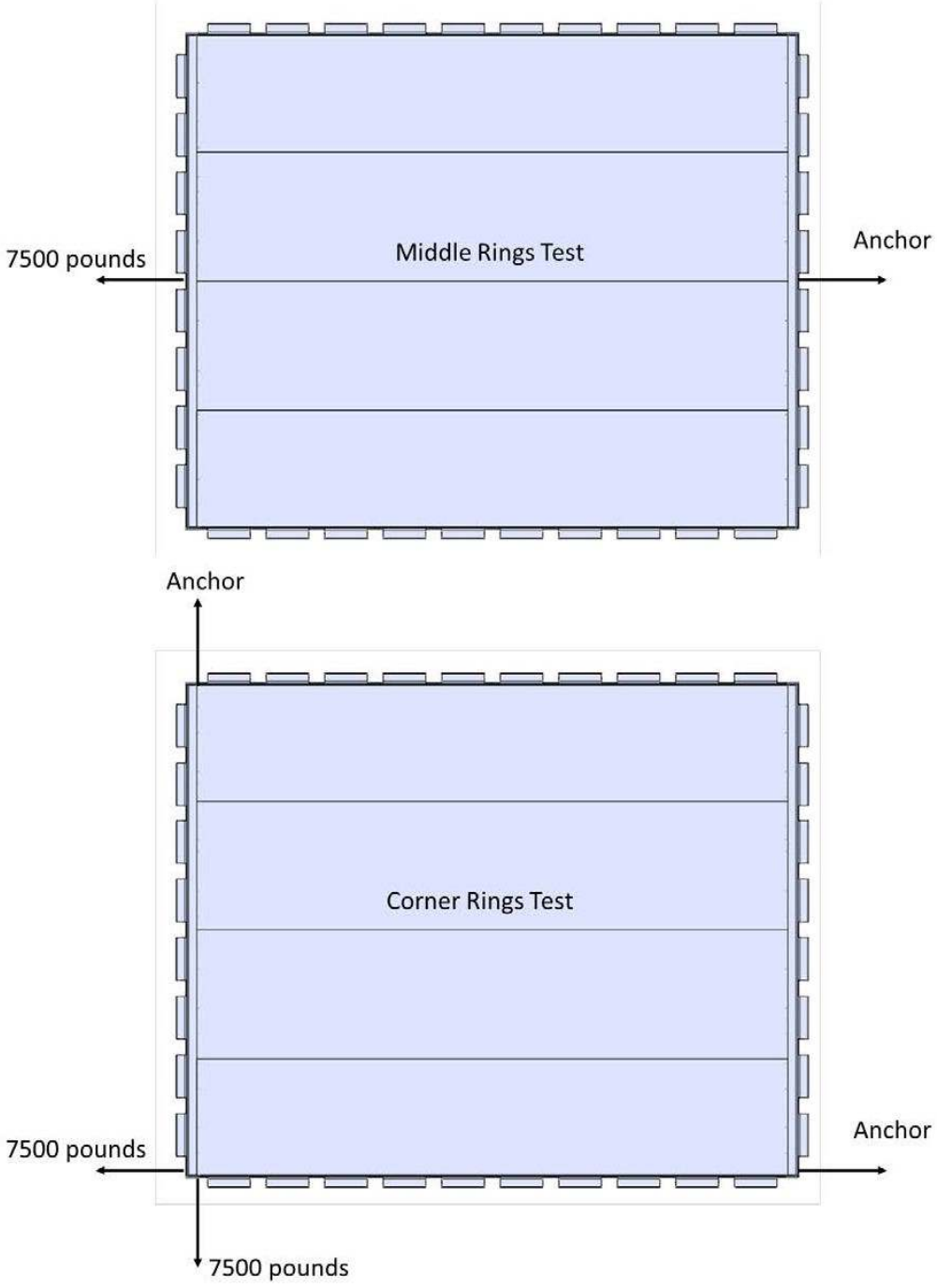


FIGURE 5. Ring tie-down test three.

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4.5.4 Static load test.
While supported by four rows of conveyors as specified (see 3.4.5), parallel to the shortest side, the pallet shall be loaded with a static load (see 6.3.2). The load shall remain on the pallet for 40 hours. The pallet shall withstand this loading condition without permanent deformation and without pallet or weld cracking (see 3.4.2).

4.5.5 Lifting capabilities tests.

4.5.5.1 Corner hoist test.
Corner hoist tests shall be conducted with a clevis or hook with a minimum cross sectional area of 0.875 square inch. The pallet shall be completely raised off the floor (see 3.4.3.1), held in that position for one minute, and then returned to the floor. This procedure shall be repeated ten times. For sample testing, the test load procedure shall be reduced on each test sample to two times. During the test, any separation of the mitered corners shall be cause for rejection. Upon completion of the test, evidence of permanent deformation, weld separation, or other weld failures shall be cause for rejection (see 3.4.3.1). Testing shall be conducted on the four corner rings on the long rails and then the four corner rings on the short rails.

4.5.5.2 Forklift lifting test.
The pallet shall be raised according to (see 3.4.3.2) to a height of 1 foot and held one minute, then lowered to rest on the conveyor. This procedure shall be repeated ten times. Upon completion of the test, evidence of permanent deformation, weld separation, or other weld failures shall be cause for rejection.

4.5.5.3 Forklift "wedge" lifting test.
The pallet shall be raised according to (see 3.4.3.3) until it clears the floor. The pallet shall then be lowered to the floor and forklift tines withdrawn. This test shall be conducted five times from each side of the pallet. Upon completion of the test, evidence of permanent deformation, weld separation, or other weld failures shall be cause for rejection.

4.5.6 Traversable capabilities tests.

4.5.6.1 Conveyor test.
In accordance with 3.4.4.1, the pallet shall be subjected to the following tests. Upon completion of the test, evidence of permanent deformation (must pass flatness requirement after test, flatness test, (see 3.5.1 and 4.5.1.1)), weld separation, or other weld failures shall be cause for rejection.

a. The pallet shall be moved a minimum of 15,000 feet in a direction parallel to the 108-inch dimension at a minimum speed of 30 feet per minute (fpm) with the loading/distance combinations shown in Table I. Any increment of movement shall be a minimum of 25 feet.
b. The pallet shall be moved a minimum of 15,000 feet in a direction parallel to the 88-inch dimension at a minimum speed of 30 fpm with the loading/distance combinations shown in Table I. Any increment of movement shall be a minimum of 25 feet.

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4.5.6.2 Conveyor ramp test.
In accordance with 3.4.4.2, the pallet shall be towed up and over a 17 degree, 10-foot long conveyor ramp connecting a single horizontal conveyor above the ramp. The pallet shall then travel back over the ramp. This procedure shall constitute one cycle. The pallet will be pulled/towed with a cable parallel to the upper horizontal conveyor that is attached to the pallet by only two rings. Attachment and winching procedures shall be in accordance with MIL-STD-1791 as shown on Figure 6. Throughout this test, the minimum towing speed of the pallet shall be 20 fpm. This test shall be repeated 30 cycles, 10 cycles in a direction parallel to the longest dimension and 20 cycles in a direction parallel to the shortest dimension. The ramps shall be constructed of roller conveyors as specified (see 3.4.4.1). Any permanent deformation (must pass flatness requirement after test, flatness test, (see 3.5.1 and 4.5.1.1)), or cracking of the pallet shall be cause for rejection.

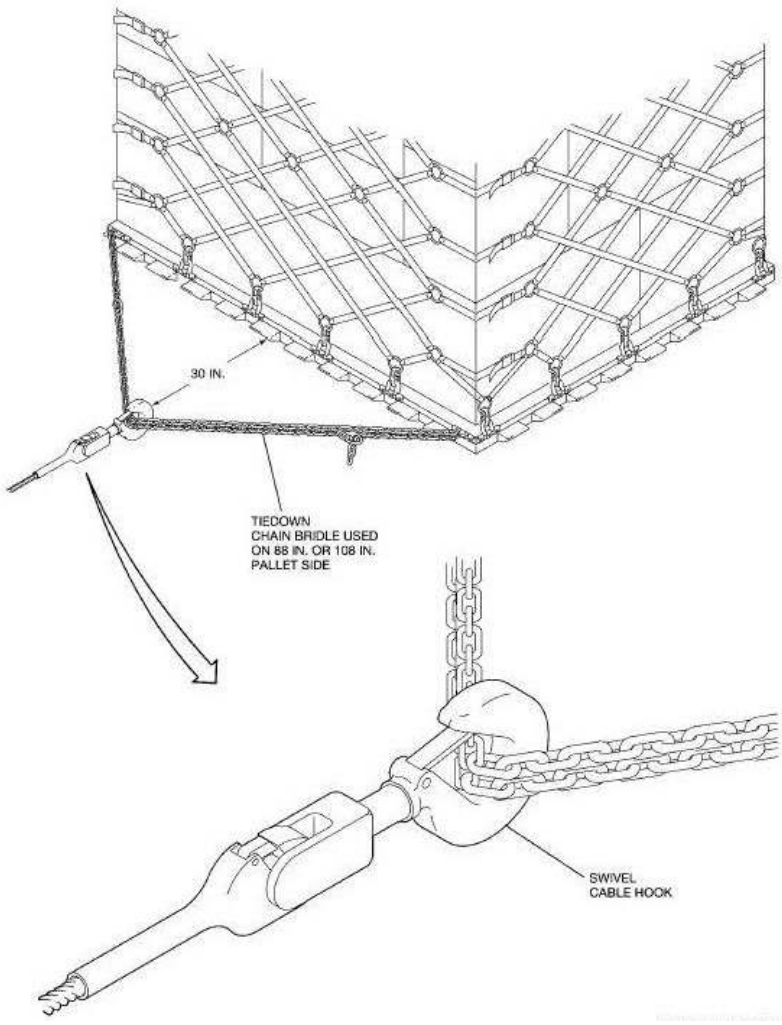


FIGURE 6. Ring attachment procedure.

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4.5.7 Ultimate load.
When uniformly loaded to rated capacity (see 6.3.1), the load being restrained to the pallet by nets (see 6.3.4), the pallet installed between restraining rails locked to the rails by two locks through each rail and engaging two lock notches on each side of the pallet, and resting on four rows of conveyor as specified (see 3.4.5), the loaded pallet shall withstand a dynamic load of three times the force of gravity (g's) for a period of 0.1 second. The pallet shall be serviceable after undergoing the test. Minimal permanent deformation is acceptable as long as the pallet will still function in a manner consistent with unloading it from an aircraft and maintain mission capabilities (ring functionality retained and passes flatness test per 3.5.1 and 4.5.1.1). In addition, the pallet shall withstand a dynamic load of 8 g's for a period of time not less than 0.1 second. The pallet need not be serviceable after undergoing such a load; however, the pallet shall remain in one piece (see 3.4.5).

4.5.8 Environmental tests.
The pallet shall be subjected to the following environmental tests in the order shown:

4.5.8.1 High temperature test.
The pallet shall be subjected to high temperature in accordance with Procedure I guidelines, Method 501.7 of MIL-STD-810. After a 24-hour soak period at 165 °F, the pallet shall be loaded with a flexible, uniformly distributed load (see 6.3.1) and subject to 50 consecutive cycles of loading as follows without permanent deformation. One cycle shall have a maximum duration of 30 seconds (see 3.4.6).

a. Lift the pallet completely off the floor by four tie-down rings located adjacent to the corners on the 88-inch sides. Lifting shall be conducted with a clevis or hook with a minimum cross sectional area of 0.875 square inch.
b. Lower the pallet to rest on three 4 × 4 × 88 inch long supports equally spaced and aligned with the short centerline. The lifting device shall be completely free of weight support while the pallet is at rest on the supports.

4.5.8.2 Water penetration and low temperature test.
Five 0.375-inch diameter holes shall be drilled through the sandwich construction. One hole shall be in the center of the pallet and two each on the transverse and longitudinal centerline 25 inches from the pallet edge. The pallet shall then be subjected to the following immersions in the sequence listed (see 3.4.6), in accordance with Method 512.6, Procedure I of MIL-STD-810:

a. Immerse 3 hours in water at room temperature; the upper surface of the pallet shall be covered by at least 2 feet of water.
b. Remove from water and weigh pallet immediately.
c. Soak in a cold chamber for 4 hours at -65 °F.
d. While the temperature remains at -65 °F, the pallet shall be subjected to 20 cycles of the same lift test specified (see 4.5.8.1.a and 4.5.8.1.b).
e. Repeat steps a through d.

The pallet shall survive the test without a structural failure or cracking.

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4.5.8.3 Humidity test.
The pallet shall withstand exposure to rain in accordance with Procedure II guidelines, Method 507.6 of MIL-STD-810. This test shall be considered qualified by similarity with current pallet, since the Next-Gen pallet is all aluminum and has eliminated the adhesives, sealants, and balsawood used in the historic 463L pallet.

4.5.8.4 Salt fog test.
The pallet shall be subjected to the salt fog test in accordance with Method 509.7, of MIL-STD-810. The duration of exposure shall be 24 hours. The test shall be repeated for four cycles (see 3.4.6). A representative portion of the pallet may be used for this test as long as it includes examples of all weld types as well as the tie-down ring assemblies. This specimen may then also be used for other testing as appropriate.

4.5.9 Destruction analysis.

4.5.9.1 Pressure/Point load test.
Each test sample shall be cut up. The specimen shall be 10 × 10 inches including three full extrusion cells, using those cells that have the thinnest outer skin dimensions. All parts of the pallet shall be permanently identified. The tests shall be as follows:

> Using a 1-inch square steel mandrel, apply a 900-pound load to the top surface of one 10 × 10 inch test panel. The load shall be applied at the mid-point of the specimen centered in the open extrusion bay (midway between vertical webs). At the option of the supplier, the corners of the mandrel may be slightly rounded. The test panel may be from either test pallet number 1 or 2. The test specimen shall withstand the 900-pound load. Permanent deformation of the test panel in the load area shall not exceed 0.005 inch in depth. Deflection while under load shall be measured, but there is no specific elastic deformation limit. (**NOTE:** this test may be conducted on a sample that is directly from an extrusion cutoff as opposed to being cut or extracted from a full pallet.)

4.5.9.2 Weld testing.
The manufacturer shall conduct FSW process development and perform welding in accordance with AWS D17.3/D17.3M, Class B (dye penetrant inspection not required). FSW testing shall be conducted on representative samples from each type of joint and weld process, using representative manufacturing processes and materials. Two FSW configurations shall be tested: the longitudinal axis joint (long weld across top/bottom joining middle panels and edge panels) and the edge rail joint (top/bottom joining the edge rails to the edge panels and middle panels). For GMAW/GTAW corner specimens, samples shall be constructed using the top and bottom diagonal joints. FSW samples shall be tested a minimum of 1-inch after start of weld and a minimum of 1-inch before end of weld. GMAW/GTAW samples shall be from the middle of the corner joint. Samples shall be removed from actual pallets, off-cuts, or representative witness samples. Test coupons shall be cut to allow qualitative inspection of weld quality (cross-sections/micrographs allowing inspection for discontinuities). Prototype manufacture shall not commence until weld qualitative assessments are complete without unacceptable defects. Note that defect and strength requirements may be relaxed if quantitative testing is conducted and approved by the government demonstrating that the proposed requirements do not result in strength reductions below those specified and do not affect other critical test capability (such as 4.5.3.4). In lieu of the AWS D17.3/D17.3M test

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coupons, test coupons in accordance with AWS B4.0, may be substituted. The test specimen shall be the 'transverse rectangular tension specimen (plate)'. Where that geometry is not possible due to pallet geometry, other suitable specimens may be substituted. For example, for GMAW/GTAW specimens from the top and bottom surface diagonal corner welds, ISO 8256, Type III specimens have been used successfully. Five specimens shall be tested for each type of weld process and for the as-extruded base material. GMAW/GTAW weld cross sections shall be inspected using the criteria from AWS D1.2/D1.2M. For GMAW/GTAW specimens, the average weld efficiency (weld specimen failure stress/parent extruded base material specimen failure stress) × 100 percent shall be greater than 33 percent. For FSW specimens, the weld efficiency (ultimate tensile strength of the welded test specimen / specified minimum tensile strength of the parent extruded base material) × 100 percent shall be greater than 60 percent.

5. PACKAGING

5.1 Packaging.

For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use.

The Next-Gen cargo pallets are intended for use in conjunction with the 463L Cargo Handling System to transport cargo in aircraft, within air freight terminals, and on motorized and non-motorized vehicles.

6.2 Acquisition requirements.

Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Packaging requirements (see 5.1).

6.3 Definitions.

For the purpose of this specification, the following definitions will apply.

6.3.1 Uniformly distributed load. A uniformly distributed load is defined as rectangular units. The total load weight for the Next-Gen pallet is 10,000 pounds.

6.3.2 Static load. The static load for the Next-Gen pallet is 45,000 pounds.

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6.3.3	Load test.	The load for the Next-Gen pallet is a uniformly distributed load of 18,000 pounds.
6.3.4	Nets.	Next-Gen pallets use two each side nets HCU-7/E or HCU-7A/E and one each HCU-15/C top net.
6.3.5	A-Serviceable.	Issuable without Qualification New, used, repaired, or reconditioned material which is serviceable and issuable to all customers without limitation or restrictions. Includes material with more than 6 months of shelf life remaining.
6.3.6	F-Unserviceable (repairable).	Economically reparable material which requires repair, overhaul, or reconditioning (includes reparable items which are radioactively contaminated).

6.4 Government-furnished property.
The contracting officer should arrange to furnish the property listed (see 3.11).

6.5 Subject term (key word) listing.

Forklift wedge
Load test
Nets
Static load
Tie-down ring
Ultimate load

6.6 International standardization agreement implementation.
This specification implements AIR-STD-25/61. When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

6.7 Changes from previous issue.
Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

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Preparing activity
Air Force - 184

(Project 1670-2019-001)

Agent:
Air Force - 11

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